

## FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

### *Student Assessment Document*

#### FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 2.6: AREA OF A PART OF A CIRCLE

##### Student Assessment Document

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| Student Name | _____ |
| Class        | _____ |
| Date         | _____ |

#### INSTRUCTIONS FOR STUDENTS

Final Explanation: Answer the Farmer's Question. Think about the Laikipia County centre-pivot irrigation farm from our lessons. The driving question is: How can we calculate the exact area of any part of a circle so that Kenyan farmers, engineers and artisans can make accurate plans and avoid waste? Answer every section in the space below it. Write each formula you use, show every step of your working, and give units in your final answers (use the  $\pi$  button on your calculator, and give answers to 2 decimal places unless told otherwise). Where you are asked to explain, write full sentences and refer to the farm, the evidence you gathered, or the models you built.

#### Part 1: Why the full-circle formula is not enough (Annulus)

The Laikipia farmer's title deed gives only the area of the full circular plot. (a) Explain why the formula  $A = \pi r^2$  alone cannot tell the farmer the area of the ring-shaped zone between the inner pipe ring and the outer pipe ring. (b) The ring lies between an inner circle of radius 25 m and an outer circle of radius 40 m, both centred on the pivot. Write the formula for the area of an annulus and calculate the area of the ring. (c) A chapati-pan border is also an annulus. Explain in one sentence how the same idea applies to it.

*Write your answer here:*

### Part 2: The pivot's sweep (Sector, arc length and perimeter)

The pivot pipe of the maize field is 40 m long, and it has rotated through  $120^\circ$ . (a) What fraction of the full circle has the pipe swept, and why does that fraction work? (b) Write the formula for the area of a sector and calculate the area of maize watered so far. (c) The farmer wants to fence the edge of this watered sector. Write the formulae for arc length and sector perimeter, and calculate the length of fencing needed.

*Write your answer here:*

### Part 3: Combining the ring and the wedge (Annular sector)

The outer irrigation ring, between radius 25 m and radius 40 m, has been swept through  $120^\circ$ . (a) Explain how the annular sector formula comes from the annulus formula and the sector formula. (b) Calculate the area of this watered wedge-shaped ring. (c) A classmate calculates  $(40 - 25)^2$  and then multiplies by  $\pi$  and by  $120/360$ . Explain what is wrong with this method.

*Write your answer here:*

### Part 4: Chord-cut regions and overlapping plots (Segment and common region)

(a) A circular Turkana borehole hatch cover has radius 35 cm. A straight locking bar (a chord) cuts off a segment whose central angle is  $110^\circ$ . Write the formula for the area of a segment and calculate the area of this segment. (b) Two circular irrigation plots on the Laikipia farm have radii  $r_1 = 42$  m and  $r_2 = 35$  m, and their centres are  $d = 49$  m apart. Explain the steps you would follow to find the area they share, including how the central angles are found. Use the cosine rule to find the central angle at the centre of the circle of radius 42 m. (c) The two segment areas are approximately  $485.8 \text{ m}^2$  and  $662.3 \text{ m}^2$ . What is the shared area, and why does it matter to the two farmers?

*Write your answer here:*

### Part 5: Answering the farmer's question

Use everything you have learned to answer the driving question: How can we calculate the exact area of any part of a circle so that Kenyan farmers, engineers and artisans can make accurate plans and avoid waste? In your answer: (a) name the five sub-regions of the Laikipia farm we studied and give the formula for each; (b) explain how you decide which formula to use for a shape; (c) give one example, other than the Laikipia farm, where a Kenyan engineer or artisan could use one of these formulae to avoid waste.

Write your answer here:

| RUBRIC                                                         |                                                                                                                                                                                       |                                                                                                                |                                                                                                                     |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Criterion                                                      | Excellent (4)                                                                                                                                                                         | Proficient (3)                                                                                                 | Developing (1–2)                                                                                                    |
| Selecting and stating the correct formula for each circle part | Names the correct formula for all five sub-regions (annulus, sector, annular sector, segment, common region) and explains why each fits the shape's structure.                        | States the correct formula for most sub-regions, with only minor slips or limited explanation of why it fits.  | States few correct formulae or confuses them, for example omitting the triangle subtraction in the segment formula. |
| Accuracy of calculations and units                             | All calculations are correct, with each radius squared before subtracting, angles used correctly, and answers given in the right units ( $\text{m}^2$ , $\text{cm}^2$ , $\text{m}$ ). | Most calculations are correct, with small rounding or unit errors that do not change the method.               | Several calculation errors, such as subtracting radii before squaring, or units missing or wrong.                   |
| Structured working and mathematical reasoning                  | Working is set out clearly step by step, and the cosine rule steps for the lens are explained and linked to the segments.                                                             | Working is mostly clear, with some steps missing or the reasoning for the common region only partly explained. | Working is hard to follow or answers are given without steps or reasons.                                            |
| Connecting to the phenomenon and Kenyan contexts               | Clearly links every formula to the Laikipia farm and gives a correct, relevant Kenyan example showing how exact area avoids waste.                                                    | Links most answers to the Laikipia farm, and gives a Kenyan example with limited explanation.                  | Little or no link to the farm or Kenyan contexts, and the answer to the driving question is incomplete.             |

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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